Real Analysis: Exercise 4

Due date:

Question 1. Let f and g be real-valued functions of a real variable. For each of the following statements, either prove it is true using the definition of the limit or give a counterexample to show that it is false.

1. Suppose that $\lim_{x\to\infty} f(x) = a$ and $\lim_{x\to\infty} g(x) = b$. If $f(x) < g(x)$ for all $x \in \mathbb{R}$, then $a < b$.
2. If $\lim_{x\to a} f(x) = \ell$ and $\lim_{x\to a} g(x) = \infty$, then $\lim_{x\to a} f(x)g(x) = \infty$.
3. If $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = \ell$, then $f(a) = \ell$.
4. If $\lim_{x\to b} f(x) = c$ and $\lim_{x\to a} g(x) = b$, then $\lim_{x\to a} f\big(g(x)\big) = c$.

Question 2. In this question you may use the Algebra of Limits but should not use l'Hôpital's rule. Evaluate the following limits.

1. $\lim_{x\to 0} \dfrac{x^2+1}{3x-5}$.
2. $\lim_{x\to 0} \dfrac{7x}{\sin(4x)}$.
3. $\lim_{x\to 2} \dfrac{\sqrt{3x-2}-\sqrt{5x-6}}{\sqrt{2x-1}-\sqrt{x+1}}$. (Consider multiplying the numerator and denominator by $\sqrt{2x-1}+\sqrt{x+1}$.)
4. $\lim_{x\to 0} \dfrac{x\sin x}{1-\cos x}$.

Question 3. Determine whether $\lim_{x\to a} f(x)$ exists and compute the limit if it exists.

$$f(x) = \begin{cases} x\sin\left(\dfrac{1}{x^2}\right), & \text{if } x \text{ is irrational,} \\ -2x, & \text{if } x \text{ is rational,} \end{cases} \qquad a = 0.$$

Question 4. Use the Squeeze (Sandwich) Theorem to evaluate

1. $\lim_{x\to 0} x^2 \cos\dfrac{1}{x}$;
2. $\lim_{x\to 0} x^4 \sin\dfrac{1}{x^2}$.

Question 5.

1. Evaluate $\lim_{x\to 0} \dfrac{\sin 3x}{\sin 5x}$.
2. Evaluate $\lim_{x\to 0} \dfrac{\tan 2x}{x}$.

Question 6.

1. Show that $\lim_{x\to 0^+} \dfrac{1}{x} = +\infty$ and $\lim_{x\to 0^-} \dfrac{1}{x} = -\infty$, and hence that $\lim_{x\to 0} \dfrac{1}{x}$ does not exist.
2. Evaluate $\lim_{x\to\infty} \dfrac{2x^2+3x-1}{x^2-4}$.